Fretboard Modeller

User guide – design a fretboard, machine it, and cut the fret slots

1. Overview

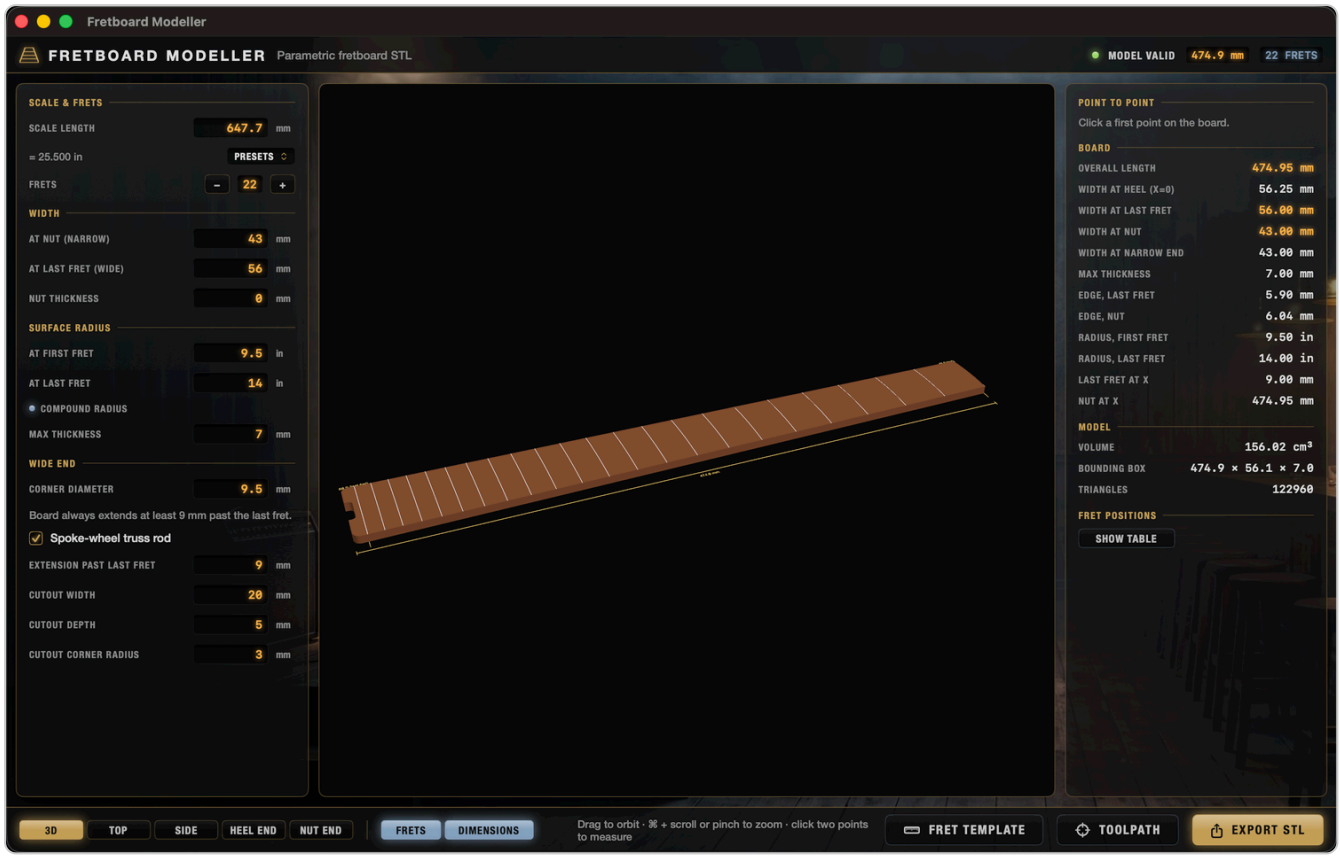
Fretboard Modeller has four tools that share one set of board dimensions:

Tool	Opened from	Produces
Designer	Tab 1 (⌘1)	3D fretboard, measurements, binary `.stl` (mm)
Markers	Tab 2 (⌘2)	Fret dots and a traced 12th-fret inlay in the model, plus their own G-code files
Toolpath	Tab 3 (⌘3)	G-code (`.nc`) for the WorkBee Z1+ or FoxAlien XE-Pro Ultra, with animated playback
Fret Template	Tab 4 (⌘4)	Marker-hole G-code for the chop-saw fret slot jig

Launch it from Spotlight (⌘ Space, “Fretboard Modeller”) after running `./make_app.sh`. The four tools are tabs along the top of the one window (⌘1 Designer, ⌘2 Markers, ⌘3 Toolpath, ⌘4 Fret Template). Toolpath is unavailable while the Designer inputs are invalid.

Projects: File › New Project (⌘N), Open… (⌘O), Save (⌘S) and Save As… (⇧⌘S) work with `.fretboard` files, which hold the board dimensions, CNC settings, markers, marker job settings and the inlay image itself. The project name is shown in the header; an amber dot (and “Edited” in the window title) means unsaved changes, and New, Open and Quit ask before discarding them. Generated toolpaths are not saved; press GENERATE again after opening.

2. Designer



Inputs

Field	Meaning
Scale length	In mm; the inch equivalent is shown, and PRESETS offers common scales.
Frets	Number of frets (1–36).
At nut / at last fret	Board width at the nut and at the last fret; the board tapers linearly between them.
Fender-style nut	Off = Gibson style, the board stops at the 0 fret. On = Fender style, the board extends 10 mm past the 0 fret (default off).
Surface radius	Radius in inches at the first and last fret. Equal values = constant radius; different values = compound radius.
Max thickness	Thickness on the centre line (7 mm default). With a compound radius the centre-line thickness stays constant and the edges get thinner toward the wide end.
Corner diameter	Rounding of the two wide-end corners (9.5 mm default). The narrow end has hard corners.
Extension past last fret	The board always extends at least 9 mm beyond the last fret.
Spoke-wheel truss rod	On by default. Adds a cutout centred on the heel edge: width 20 mm, depth 5 mm, inside corner radius 3 mm (all editable).

Orientation

The widest end (heel) sits at X = 0; the narrowest end points toward +X. Y is across the board, centred on 0. Z is up, with the bottom of the board at Z = 0.

Viewer and measurements

- Drag to orbit; ⌘ + scroll or pinch (Magic Mouse / trackpad) to zoom.
- Buttons along the bottom set 3D, TOP, SIDE, HEEL END and NUT END views and toggle fret lines and dimension lines.
- Click two points on the board for a point-to-point distance.
- The right panel shows overall length, widths, edge thicknesses, radii, volume, bounding box and triangle count; SHOW TABLE lists every fret position.

Exporting

EXPORT STL writes a binary STL in millimetres. The mesh is checked to be watertight before export is allowed; if a setting is invalid the status badge turns red and tells you why.

Markers

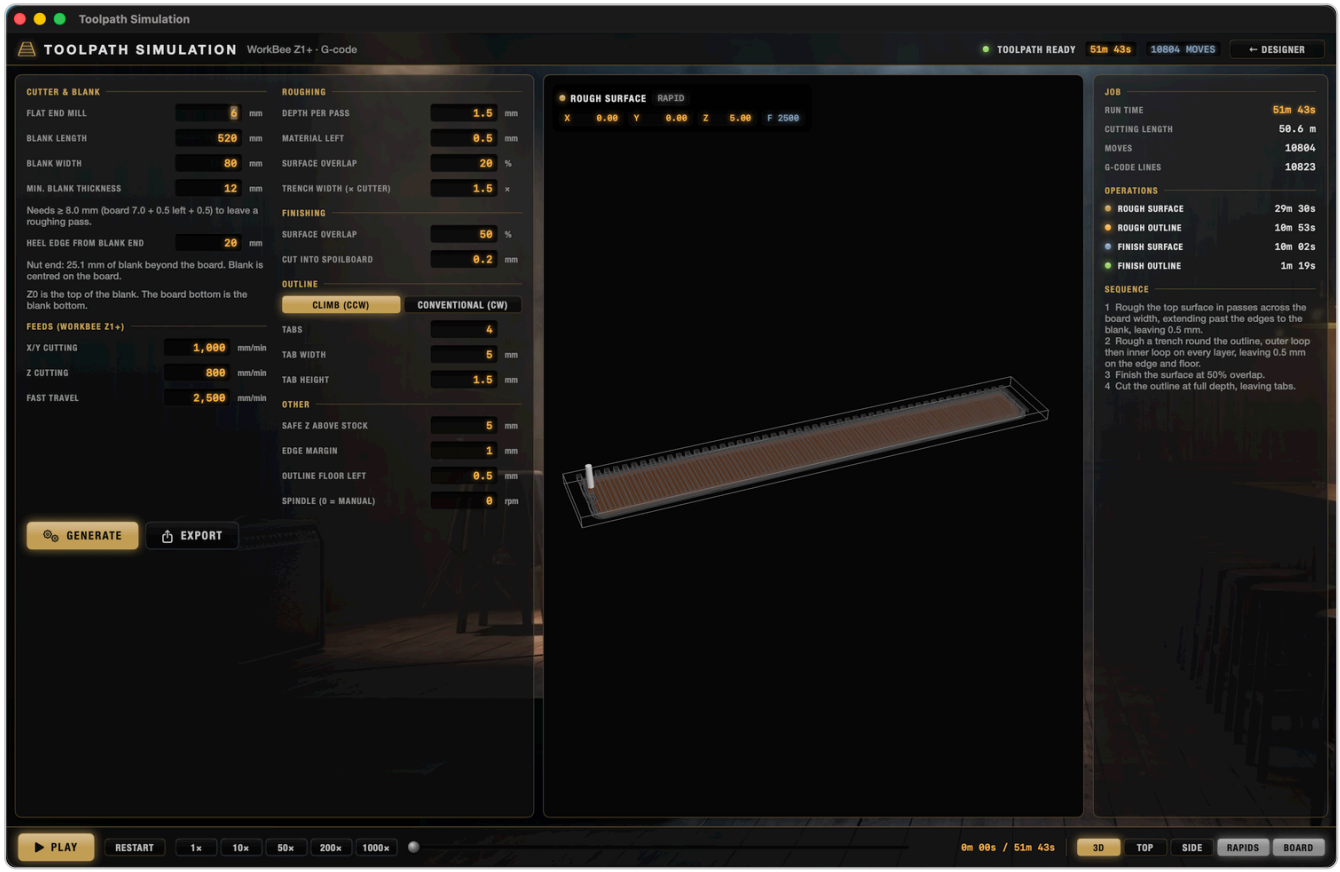
Choose **Dots**, **Dots + 12th inlay** or **12th inlay only**. Dots are 6 mm × 3 mm in the middle of the selected fret spaces; the 12th and 24th are double dots under the B and A strings, using compensated string spacing from the nut (E strings 3 mm in from the edges) to a 52.4 mm tune-o-matic. The inlay starts from the bundled default design, 45 mm tall across the board; to use your own, **IMPORT** an image, **PASTE** one, drop one on the preview, or **CAPTURE** an area of the screen (allow Screen Recording if asked). Black becomes the pocket. Adjust the trace, then size (aspect locked or free), move, rotate or mirror it. The pockets appear in the Designer model and STL; the inlay pocket is the shape the inlay cutter actually cuts, so parts too narrow for it are left standing, as they will be on the board. The board is right-handed by default (bass side at +Y); the plan is drawn the same way round as the Designer's top view (nut on the right, bass side at the top), and the inlay reads upright there. Tick *Left-handed board* for a left-handed neck.

Press **GENERATE** and export the two G-code files. The *dots* file uses the board job's origin and the 6 mm cutter. The *inlay* file is for the fine cutter and can use the board origin or the centre of the inlay (Z0 on the board surface there). It ramps into every loop, slows down in full-width slots, and finishes the walls lightly on each layer. Red areas in the preview are narrower than the cutter and will be left; they don't count toward the 2 mm edge clearance. If you change the markers, the image or a job setting after generating, an **UPDATE G-CODE** prompt appears, and exporting out-of-date G-code asks first. If the markers can't be cut as they are, the reason is shown under GENERATE.

After generating, switch the preview from **PLAN** to **3D DOTS** or **3D INLAY** to play a job on the board with a moving cutter and live X/Y/Z/feed readout, using the same controls as the Toolpath tab. The readout is in the job's own coordinates, so it matches the file even with the inlay-centre zero.

> Run marker jobs after surfacing and before releasing the board from its tabs, with the same work offsets. Air-cut first with Z raised.

3. Toolpath (CNC)



Open the **TOOLPATH** tab. Set the blank and options on the left, press **GENERATE**, review the run in the viewer, then **EXPORT** to save the G-code. The status badge shows *OUT OF DATE* if you change the board after generating – generate again before exporting.

Coordinate system

X0 = heel edge of the board, Y0 = board centre line, **Z0 = top of the blank**. The bottom of the board is the bottom of the blank. Cutter: 6 mm flat end mill. Set these work offsets on the machine before running.

Blank

Setting	Notes
Blank length / width	Defaults 520 × 80 mm. The blank is centred on the board's centre line.
Min. blank thickness	Default 12 mm. Must be at least board thickness + material left + 0.5 mm (8.0 mm for the default 7 mm board) so there is at least one roughing pass. Thickness blanks beforehand to about this value.
Heel edge from blank end	Default 20 mm; the nut-end margin is calculated and shown.

> The path is clipped to the blank and the app rejects a blank that cannot contain the board (1 mm all round). An amber warning appears if the blank is too tight for the outline trench.

Sequence

1. **Rough surface** – raster passes across the board width, running past the board edges, 1.5 mm deep, leaving 0.5 mm, 20 % overlap.
2. **Rough outline** – a trench round the board 1.5× the cutter width wide, cut as two loops on every depth layer: the outer loop first and the loop beside the board last, so the board edge gets the lightest cut. Leaves 0.5 mm on the edge and floor.
3. **Finish surface** – 50 % overlap, removing the last 0.5 mm.
4. **Finish outline** – full-depth cut round the board, leaving tabs.

The order (surfaces and trench first, final outline last) is chosen to reduce splitting.

Tabs and outline direction

- **Tabs** hold the board in the blank on the final cut: default 4, each 5 mm wide and 1.5 mm tall. Cut them away by hand afterwards. Set tabs to 0 if you hold the blank down some other way (tape and superglue work as double-sided tape).
- **Climb (CCW)** or **Conventional (CW)** sets the outline direction. If you see splits on one side of the board, switch direction.

Feeds

X/Y cutting 1000 mm/min, Z cutting 800 mm/min. **Fast travel defaults to 2500 mm/min as a placeholder – set it to your machine's configured maximum.** Spindle RPM 0 means manual spindle control (no M3/M5 written).

Playback

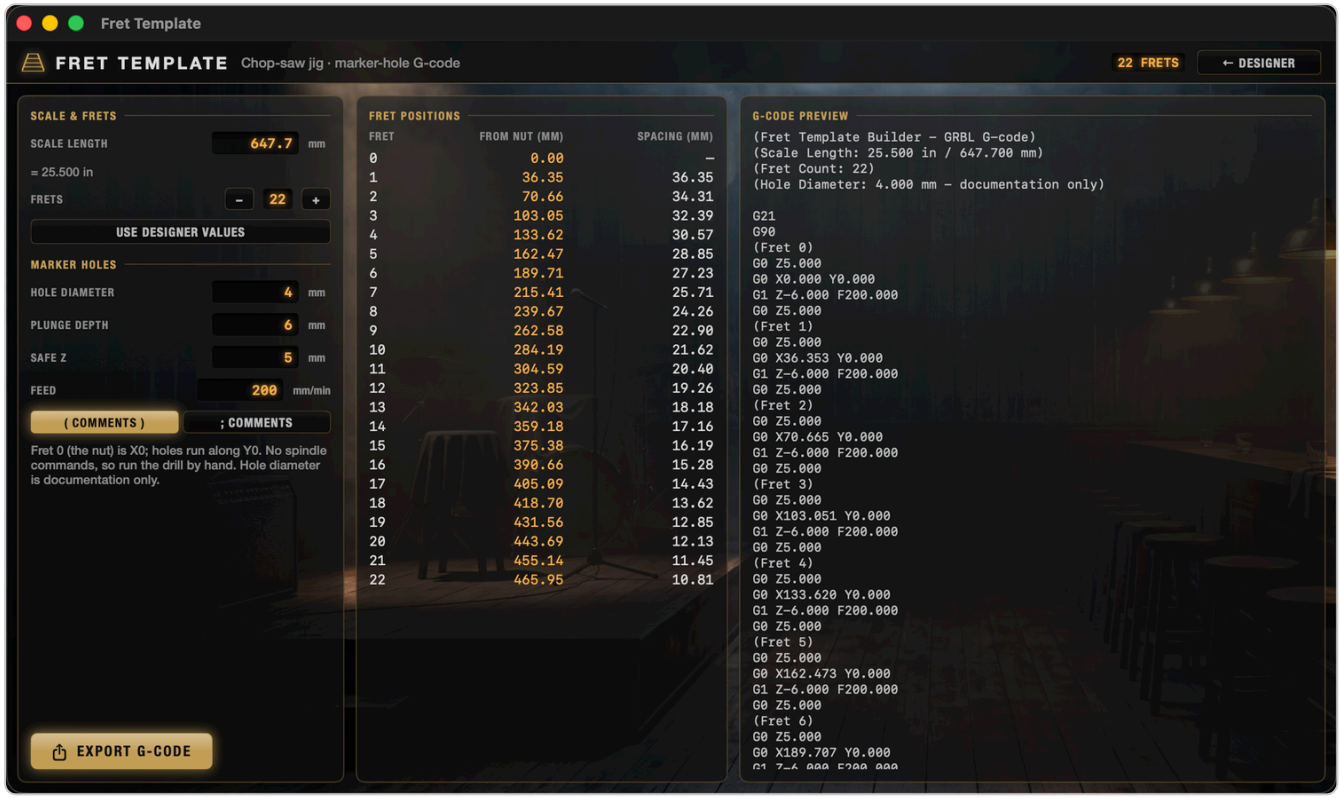
Play the run at 1×, 10×, 50×, 200× or 1000× with the slider; the cutter and live X/Y/Z/feed readout follow the path. Views: 3D / TOP / SIDE; toggle RAPIDS and BOARD. Path colours: gold = rough surface, amber = rough outline, blue = finish surface, green = finish outline. Playback shows the cutter path, not material removal.

G-code format

Pick the machine in the Toolpath tab: **Ooznest WorkBee Z1+** (Duet / RepRapFirmware) or **FoxAlien XE-Pro Ultra** (GRBL). Choosing one loads its starting feeds and spindle speed, which you can change. The FoxAlien output uses `G94` and `G54`, `M3 S<rpm>` followed by a 5 second `G4` spin-up for the VFD spindle, `(parenthesis)` comments on their own lines, no feed word on rapids, and ends with `M5` and `M2`. It runs from USB or the offline controller.

WorkBee / Duet / RepRapFirmware: `G21`, `G90`, `G17`, explicit `G0` / `G1` with feed rates, header comments describing the job, and a section comment per operation. Check the output against your machine's configuration before the first run, and test above the stock.

4. Fret Template



Creates the drilled-hole template used to position a jig for cutting fret slots on a chop saw. Fret positions come from $d = L \cdot (1 - 1/2^{n/12})$, measured from the nut. Fret 0 is X0 and all holes lie on Y0. Tick **Fender-style nut** (it follows the Designer) to add fret –1, a second slot 4 mm past the nut at X–4.

Setting	Default
Scale length / frets	Taken from the designer; **USE DESIGNER VALUES** resets them
Hole diameter	4 mm (written as a comment only – drill with a 4 mm bit)
Plunge depth	6 mm
Safe Z	5 mm
Feed	200 mm/min
Comments	( ) or ; style

The table lists each fret's distance from the nut and spacing from the previous fret; the preview shows the G-code. **EXPORT G-CODE** saves `fret-slot-template-<scale in>-<frets>.nc`. The output is GRBL-style (ends with `M30`) and has no spindle commands, so run the drill by hand. Holes are at the fret centrelines; no blade-kerf offset is applied.

5. Building and testing

`./make_app.sh`	Builds the release app and installs it to ~/Applications (Spotlight-launchable).
`swift run FretboardApp`	Runs from the command line.
`swift test`	Runs the test suite (see the README for the Command-Line-Tools-only flags).

6. Troubleshooting

Symptom	Fix
Status badge red in the designer	Read the message; e.g. heel narrower than nut, radius too small for the width, or cutout too big.
Toolpath says blank is too small / too thin	Increase the blank size or thickness; see the minimum shown under Min. blank thickness.
Toolpath “out of date”	Press GENERATE after changing the board or settings.
Splitting on one edge	Switch the outline direction (climb / conventional).
Travel feels slow or too fast	Set Fast travel to your machine's real maximum.